1(3) 2(2) 3(4) 5(2) 8(1, 2) 9. 11

#1 (3)

$\left(2, -\frac{2}{3}\pi\right)$

$x = 2\cos\left(-\frac{2}{3}\pi\right) = 2\cdot\left(-\frac{1}{2}\right) = -1$

$y = 2\sin\left(-\frac{2}{3}\pi\right) = 2\cdot\left(-\frac{\sqrt{3}}{2}\right) = -\sqrt{3}$

$\therefore (-1, -\sqrt{3})$

#2 (2)

$\frac{3}{2}\pi +$

$(2\sqrt{3}, -2)$ : 4사분면

$r = \sqrt{12+4} = 4$ "

$\tan\theta = -\frac{2}{2\sqrt{3}} = -\frac{\sqrt{3}}{3}$ "

$\therefore \left(4, \frac{11}{6}\pi\right)$

116
330
180
6

#3 (4)

$x^2 + (y+4)^2 = r^2$

$16 = r^2$

$\therefore r = 4$

$\Rightarrow x^2 + (y+4)^2 = 16$

$x = r\cos\theta$, $y = r\sin\theta$, $x^2+y^2 = r^2$ 대입

$x^2+y^2+8y = 0$

$r^2\cos^2\theta + r^2\sin^2\theta + 8r\sin\theta = 0$

$\therefore r = -8\sin\theta$ "

#5 (2)

$x = r\cos\theta$, $y = r\sin\theta$, $x^2+y^2 = r^2$ " 이용

$r = \frac{3}{1+\cos\theta}$

$r + r\cos\theta = 3$

$\sqrt{x^2+y^2} + x = 3$

$\sqrt{x^2+y^2} = 3 - x$ ( )²

$x^2 + y^2 = x^2 - 6x + 9$

$\therefore y^2 + 6x = 9$

#8 (1)

$r = a(1+\cos\theta)$ $(a>0)$

$r^2 = ar + ar\cos\theta$

$x^2+y^2 = a\left(\sqrt{x^2+y^2} + x\right)$ "

미분 $\Rightarrow 2x + 2y\cdot y' = a\left(\frac{2x+2y\cdot y'}{2\sqrt{x^2+y^2}} + 1\right)$ "

$y' = 0$ 대입,

$2x = a\cdot\left(\frac{2x}{2\sqrt{x^2+y^2}} + 1\right)$

$2\cdot r\cos\theta = a\left(\frac{r\cos\theta}{r} + 1\right)$

$= 2r\cos\theta - a\cos\theta = a$

$\cos\theta = \frac{r}{a} - 1$ 대입

$2r\left(\frac{r}{a} - 1\right) - a\cdot\left(\frac{r}{a} - 1\right) = a$

$\frac{2r^2}{a} - 2r - r = 0$

$3a = 2r$ $\therefore r = \frac{3}{2}a$, $\cos\theta = \frac{1}{2}$

$\therefore (r, \theta) = \left(\frac{3}{2}a, \pm\frac{\pi}{3}\right)$